Bulk solids, surfaces and vessel constraints

Optional background for the independent ECS replacement lesson. Read as needed; all required calculations, Lab work and reflection are already included in the lesson.

What fixes the gas pressure when a bulk solid can appear or disappear?

Three different gas-solid questions

Bulk solid-vapor equilibrium transfers the same substance between phases, as in sublimation. Adsorption describes uptake at a surface. A gas-solid reaction changes chemical species and needs reaction stoichiometry and an equilibrium constant. Name the process before choosing equations. Lab 10 treats nonreacting bulk equilibrium; Lab 11 treats adsorption. The current reaction-extent solver does not solve general heterogeneous reactions.

The vessel matters

For a rigid closed vessel at fixed T and V, equilibrium minimizes Helmholtz energy under the material constraints. For fixed T and P, the corresponding potential is Gibbs energy. In the ideal-gas sublimation model, solid present at equilibrium fixes the subliming species partial pressure at Psub(T). There must be enough total material to fill the gas at that pressure.

Depletion and other equilibria

If the available amount is below the saturated gas capacity, the solid disappears and the gas is undersaturated. Inert gas contributes to total pressure in this ideal model. For SLE, identify the solid standard state and liquid activity. A bulk solubility condition and an adsorption loading equation describe different phase inventories.

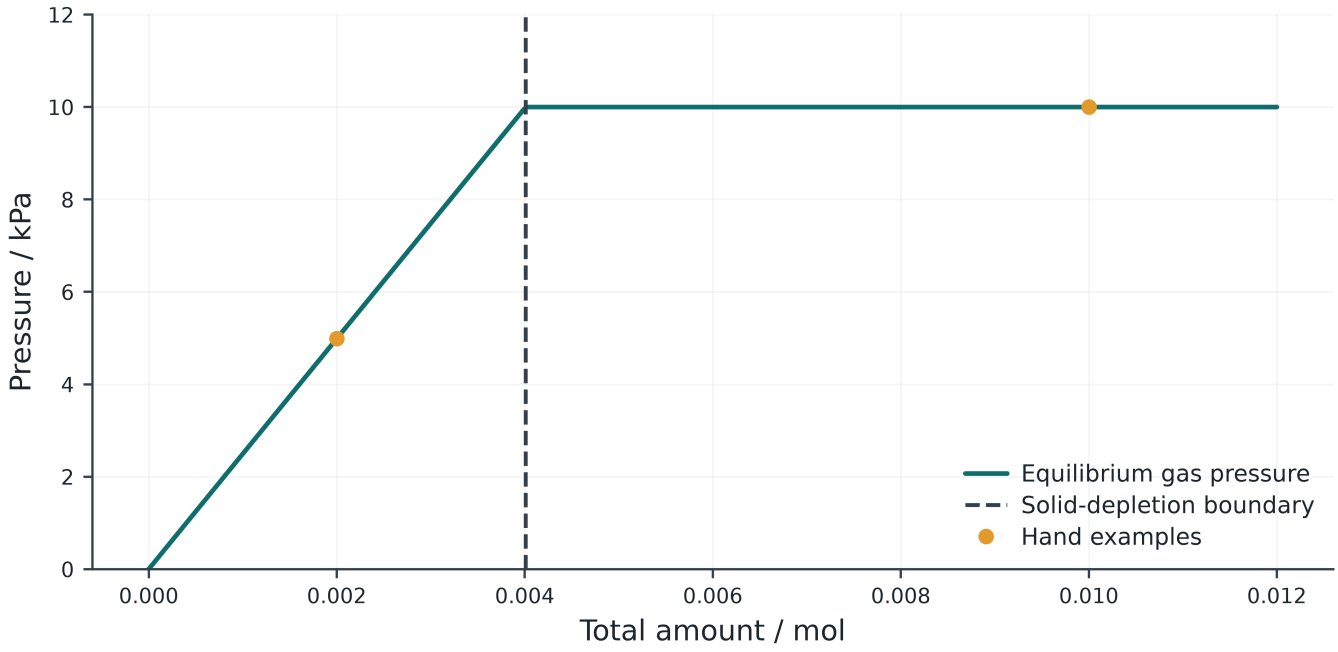
$n_{g,sat} = P_{sub} V/(RT)$; $n_g + n_s = n_{total}$

$P = n_{total} RT/V$ when the solid is exhausted (no inert gas)

One example to reason through

Synthetic teaching values. These numbers illustrate the assumptions; they are not measurements of a real material.

Assume a synthetic material at 300 K with Psub=10.0 kPa in a 1.00 L gas volume. Use R=8.314462618 kPa L mol−1 K−1. Saturated gas requires 0.00400908 mol. With 0.0100 mol total, the gas remains at 10.0 kPa and 0.00599092 mol is solid. With only 0.0020 mol, all material is gas and P=4.98868 kPa.



Check and interpret

For a separate Langmuir teaching example, qm=2.00 mmol/g and b=0.100 kPa−1 give q=qm bP/(1+bP)=1.00 mmol/g at 10 kPa. To close a vessel balance, multiply loading by adsorbent mass and convert mmol to mol. This surface amount is not the bulk solid remaining in the sublimation calculation.

Optional self-checks

If you use this optional reading, sketch or write a short prediction before opening the calculator. Revisit it after the demonstration or your own Lab calculation.

1. Classify sublimation, adsorption and a gas-solid reaction by what changes physically.

2. Predict pressure when the total amount increases from 0.002 to 0.010 mol in the example.

3. Which thermodynamic potential matches a rigid closed vessel at fixed temperature?

Optional extension

Henry gas-solubility exercise: using the explicitly defined convention p=H x with H=1000 kPa, find x for p=10 kPa. Distinguish this H from a Henry adsorption coefficient. Osmosis exercise: for an ideal dilute nonelectrolyte, estimate $\Pi = cRT$ at $c = 10\ \mathrm{mol/m^3}$ and T=300 K. These are paper extensions, not implemented new lab modes.

Continue in the core lab

https://www.skhgroup.net/teaching/2105603/labs/solids.html

Reading locator: supplied textbook contents, sections 16.1, 16.5, 16.8-16.9; compare 17.5.

Original course explanation and examples, not a reproduction of the textbook. Section numbers are based on the supplied contents pages. This reading and its extensions are optional. All required Week 4 work is included in the approximately three-hour independent lesson; no extra preparation or follow-up assignment is required.